Feature Engg. (Continued)

* Feature Scaling :-

I. Min-Max Scaling :- [It is also known as Normalization]

Formula →

$$x' = \frac{x - x_{min}}{x_{max} - x_{min}}$$

where, x = data point
x_{max} = max value
x_{min} = min value

Ex- Data → 10, 20, 25, 40

$x_{max} = 40$, $x_{min} = 10$

① $x_1 = 10$

$$x_1' = \frac{x_1 - x_{min}}{x_{max} - x_{min}}$$

$$x_1' = \frac{10 - 10}{40 - 10}$$

$$x_1' = 0$$

② $x_2 = 20$

$$x_2' = \frac{20 - 10}{40 - 10}$$

$$x_2' = \frac{10}{30}$$

$$x_2' = 0.33$$

③ $x_3 = 25$

$$x_3' = \frac{25 - 10}{40 - 10}$$

$$x_3' = \frac{15}{30} = 0.5$$

④ $x_4 = 40$

$$x_4' = \frac{40 - 10}{40 - 10} = \frac{30}{30}$$

$$x_4' = 1$$

Min-Max Scaling is a technique where we scale down the values in a range of 0 to 1

Limitation :-

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It cannot be used in case of outliers (skewed data)

II. Standard Scaling :- [It is also known as standardization]

Formula :-

$$x' = \frac{x - \mu}{\sigma}$$

where, x = data point
μ = mean value
σ = std.

Let us assume → 23, 28, 35, 18, 15, ... & $\mu = 27.1$, $\sigma = 8.66$

$x_1 = 23$

$$x_1' = \frac{x_1 - \mu}{\sigma}$$

$$x_1' = \frac{23 - 27.1}{8.66}$$

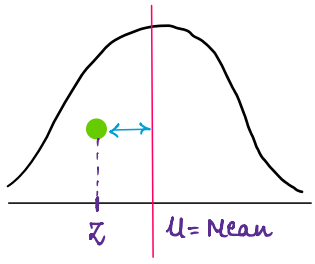
$$x_1' = -0.47$$

- Here, we can observe -ve values as well.
- It can be used in case of outliers
- The values here fall within a range of -3 to +3
- After applying standard scaling, the data forms a standard normal distribution

This $\frac{x-\mu}{\sigma}$ is also known as Z-score value.

$$Z = \frac{x-\mu}{\sigma}$$

Z-score measures how many standard deviations away is a point from its mean.



Z values generally lies in b/w -3 to +3

* Outlier Detection & Treatment :-

Outliers → Extreme values in our dataset
- ① Very small
- ② Very big

How to detect outliers ?

① Z-score Method

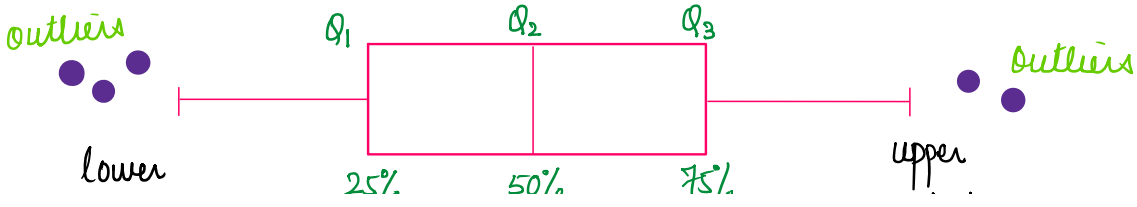
② IQR Method

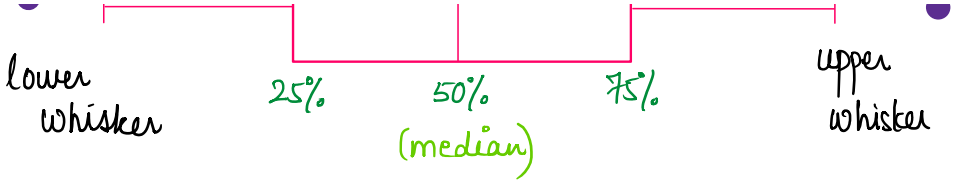
IQR :- Interquartile Range

In IQR, we calculate quartiles → Q_1, Q_2, Q_3

$Q_1 = 25\%$, $Q_2 = 50\%$, $Q_3 = 75\%$

To detect outliers → we use boxplot





Ex $-7, -5, 0, 2, 6, 9, 13, 14, 18, 25, 79$

Q_1 Q_2 Q_3

$Q_1 = 0$ & $Q_3 = 18$

Now,

$$IQR = Q_3 - Q_1$$
$$= 18 - 0$$
$$= 18$$

$$\text{upper whisker} = Q_3 + 1.5 * IQR$$
$$= 18 + 1.5 * 18$$
$$= 45$$

$$\text{lower whisker} = Q_1 - 1.5 * IQR$$
$$= 0 - 1.5 * 18$$
$$= -27$$

On seeing the above data & comparing → only one outlier point is detected, i.e., 79

↓

How to treat this outlier?

↙

For treating the outliers, we can either remove them or we can replace them.

How? → with median

In case of outliers, mean does not give an appropriate value

In case of outliers, mean does not give an appropriate value

Ex-

4000 5000 7000 8500 9000 7000000 → outlier

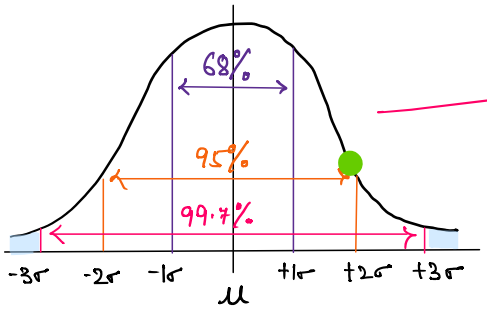
Mean = 1172250 → not appropriate

Similarly in missing value treatment also, we use median to fill the null cells in case of outliers

* Empirical Rule :-

- also known as 68-95-99.7 rule
- It is a rule that is applied to Normal Distribution
- It tells us how much data lies within certain no. of standard deviations from the mean.

Ex-



→ symmetric

for a normal distribution :-

(1) 68% of the data lie within 1 std of the mean

$$\mu - \sigma \leq X \leq \mu + \sigma$$

(2) 95% of the data lie within 2 std. of the mean

$$\mu - 2\sigma \leq X \leq \mu + 2\sigma$$

(3) 99.7% of the data lie with 3 std. of the mean

$$\mu - 3\sigma \leq X \leq \mu + 3\sigma$$

NOTE:- If $X < \mu - 3\sigma$ or $X > \mu + 3\sigma$

then $X \rightarrow$ outlier